Azure Bastion vs Jump Server

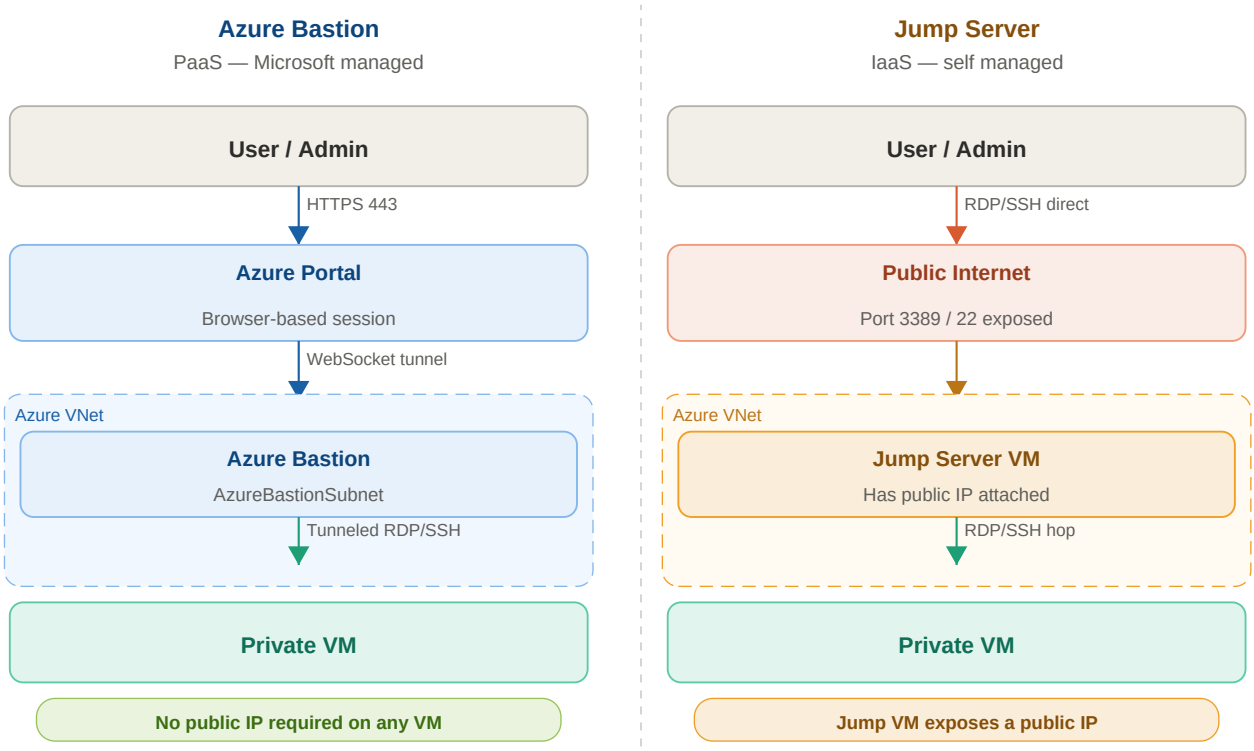
Visual Architecture Guide

DevOps · Azure · Interviews · Architecture Discussions

Senior Azure Cloud Architect & DevOps Expert

Covers: Architecture · Comparison · Scenarios · Decision Tree · Setup · Security

1 · Architecture Flow Diagrams



Left: Azure Bastion — user connects via browser over HTTPS. No RDP/SSH client needed, no public IP on target VMs. Right: Jump Server — user opens an RDP/SSH client to a public IP on the Jump VM, then hops to private VMs.

2 · Side-by-Side Comparison

Attribute	Azure Bastion	Jump Server
Type	PaaS — Microsoft-managed service	IaaS — VM you provision and manage
Security	HIGH — no public IP on any VM; encrypted tunnel	MEDIUM — Jump VM has public IP exposed
Cost	Higher (~$140–$300/mo, Standard SKU)	Lower (~$30–$100/mo, VM + license)
Maintenance	None — Microsoft patches and operates	Manual — you patch OS, monitor uptime
Access method	Browser → Azure Portal (no client app needed)	RDP/SSH client → Jump VM IP → target VM
Public IP on target VMs	Not required	Not required
Public IP on gateway	Azure-managed, transparent	Required on Jump VM — your responsibility
MFA / Entra ID	Native — Azure AD + Conditional Access	Manual — requires extra configuration
Session recording	Built-in — Bastion Insights + diag logs	Not included — needs 3rd-party tooling
Scaling	Automatic — Microsoft manages capacity	Manual — add more Jump VMs as needed
Best use cases	Enterprise, regulated industries, zero trust	Dev/test, multi-cloud, budget-constrained

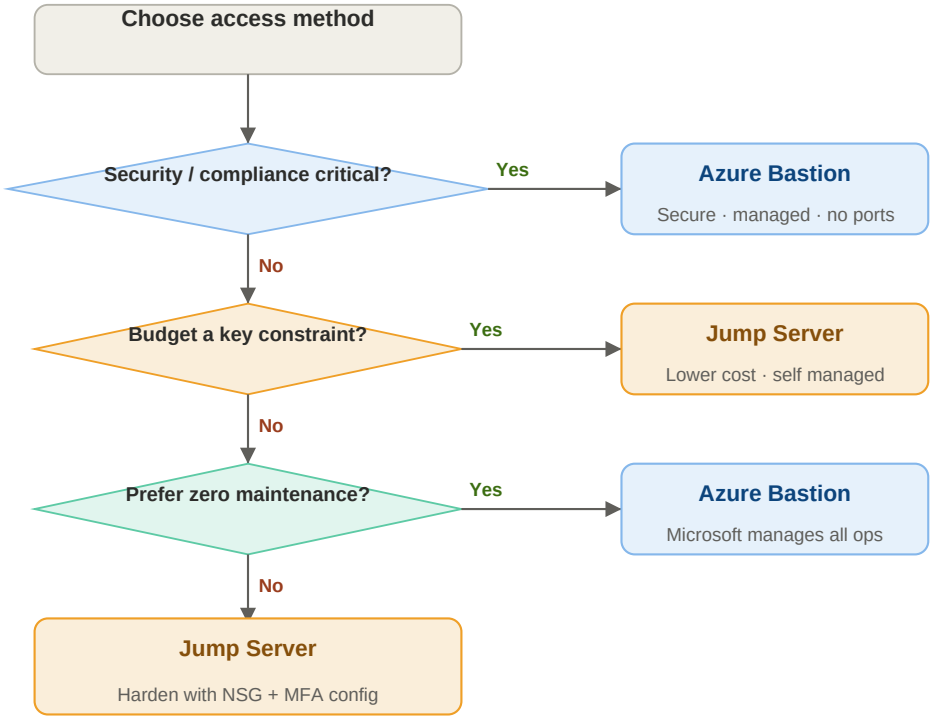
3 · Layered Architecture

Layer	Azure Bastion	Jump Server
Internet / User	Web browser — HTTPS only, no VPN, no client app	RDP / SSH client via public internet
Public access	Azure Portal URL → Bastion blade → Connect button	Public IP of Jump VM directly reachable
Gateway (★ Key difference)	Azure Bastion (PaaS) AzureBastionSubnet (/27+) Fully managed · HA built-in	Jump Server VM (IaaS) Any subnet · Windows or Linux You manage completely
VNet	AzureBastionSubnet + target VM subnet (same VNet)	Jump VM subnet + target subnet (same or peered)
VM layer	Private VMs — no public IP on any target	Private VMs — only Jump VM needs public IP

4 · Real-World Scenarios

Scenario	Recommended	Reason
Bank / financial firm (SOC2, PCI-DSS, HIPAA)	**Azure Bastion**	MFA, audit logs, no exposed ports
Dev/test startup with tight budget	**Jump Server**	Lower cost, quick to set up
Enterprise with 50+ private VMs across subscriptions	**Azure Bastion**	Scalable, centralised access
Multi-cloud / hybrid (Azure + AWS / GCP)	**Jump Server**	Cloud-agnostic, flexible
Zero trust security architecture	**Azure Bastion**	No open ports, native Entra ID
Rapid POC, demo, or temporary access	**Jump Server**	Fast deploy, easy teardown

5 · Decision Flowchart



Follow the diamonds top to bottom. First match wins — answer 'Yes' to jump to the result on the right; 'No' drops you to the next question.

6 · Step-by-Step Configuration Flows

Azure Bastion Setup

1	**Create VNet + /27 subnet** Name it 'AzureBastionSubnet' — this exact name is required
2	**Deploy Bastion resource** Portal → search 'Bastion' → Create → select VNet
3	**Provisioning (~10 min)** Microsoft deploys and configures the Bastion host
4	**Open target VM blade** VM → Connect → Bastion tab → fill credentials
5	**Authenticate in browser** Full RDP/SSH session runs inside your browser tab

Jump Server Setup

1	**Create VNet + subnet** Any subnet name works — standard networking setup
2	**Deploy VM + public IP** Create a Windows/Linux VM, attach a static public IP
3	**Configure NSG rules** Allow inbound port 3389 (RDP) or 22 (SSH) from your IP only
4	**RDP/SSH to Jump VM** Use your client to connect to the Jump VM public IP
5	**Hop to private VMs** From Jump VM, RDP/SSH internally to target VMs using private IPs

7 · Security Deep Dive

Key insight: Bastion eliminates the attack surface entirely — no ports open on target VMs, no public exposure. Jump Server shifts risk from target VMs onto the Jump VM itself, which you must harden manually.

Security Attribute	Azure Bastion	Jump Server
Attack surface	MINIMAL — no RDP/SSH ports exposed to internet	HIGHER — public IP exposes at least one VM
Port exposure	None — only HTTPS 443 inbound to Bastion	High — port 3389/22 open to internet
Brute force risk	None — no credentials sent over public ports	High — public RDP/SSH is common attack vector
MFA / Authentication	Native — Azure Entra ID + Conditional Access	Manual — must configure separately
Session recording	Built-in — Bastion Insights + diagnostic logs	Not included — needs 3rd-party tooling
OS patching	Microsoft — automatic, no action required	Manual — you own all OS patches
Zero trust readiness	Native — no lateral movement, least privilege	Configurable — requires extra policy + tooling
NSG management	Single standardised NSG for AzureBastionSubnet	NSG on Jump VM + each target subnet — complex
Compliance certs	Ready — PCI-DSS, HIPAA, SOC2 alignment built-in	Manual — must document and configure separately

Quick Reference — Interview Cheat Sheet

Azure Bastion	Jump Server
PaaS — Microsoft manages everything	IaaS — you manage everything
Connect via browser, no RDP client needed	Connect via RDP/SSH client to public IP
No public IP on any target VM	Jump VM must have a public IP
Native MFA via Entra ID	MFA requires manual config
Higher cost (~$140–$300/mo)	Lower cost (~$30–$100/mo)
Zero maintenance, automatic patching	Manual patching, hardening required
Best: enterprise, compliance, zero trust	Best: dev/test, multi-cloud, tight budget

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